Chp-1

Signal →

A signal is a function representing a variable or physical quantity, and contains information about the behaviour or nature of the phenomenon.

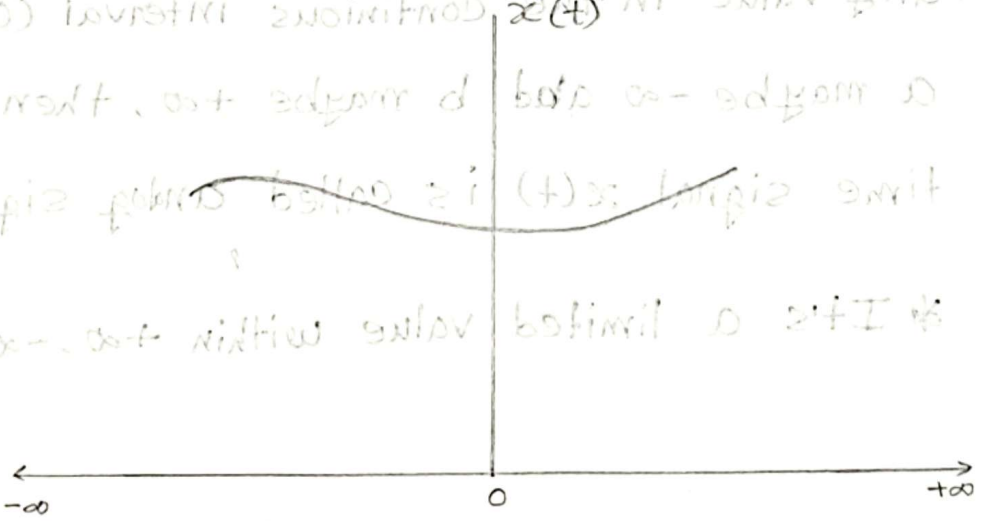
Classification →

A. Continious-Time signals

Its a signal where 't' is a continious variable.

Signal represent infinite number.

Function → $x(t)$; t = time-domain (input).

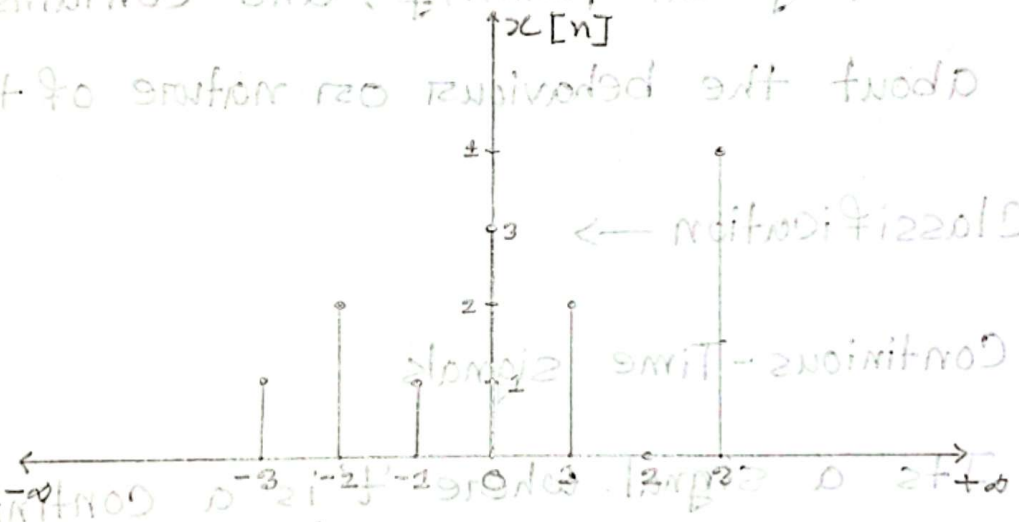


B. Discrete-Time signals —>

It's a signal where 'n' is a integer number.

Signal represent finite number.

Function —> $x[n]$; n = integer.



C. Analog signals —>

If a continious-time signal $x(t)$ can take on any value in the continious interval (a,b), where a maybe $-\infty$ and b maybe $+\infty$, then continious-time signal $x(t)$ is called analog signal.

* It's a limited value within $+\infty, -\infty$.

D. Digital signals →

A digital signal is a discrete-time signal that can take any values only from a finite set of distinct levels. $x[n]$ can take any values from a finite set.

* It has a limit value within finite set.

E. Real signals →

A signal $x(t)$ is a real signal if its value is a real number.

F. Complex signals →

A signal $x(t)$ is a complex signal if its value is a complex number.

G. Deterministic signals →

Fully predictable, no randomness.

H. Random signals →

Cannot be predicted exactly.

* $x(t)$ = noise process.

I. Even signals →

* $x(-t) = x(t)$

* $x[-n] = x[n]$

J. Odd signals →

* $x(-t) = -x(t)$

* $x[-n] = -x[n]$

K. Periodic signals →

A continious time signal,

* $x(t) = x(t+T) = x(t+mT)$

A discrete time signal,

* $x[n] = x[n+T] = x[n+mT]$

Signal that repeats after a fixed interval or time period T.

L. Non-periodic signals →

Do not repeat over time.

M. Energy signals →

$$* E = \int_{-\infty}^{\infty} |x(t)|^2 \cdot dt$$

$x(t)$ = continious time signals

N. Power signals →

$$* P = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \cdot d$$

Continious-Time signal

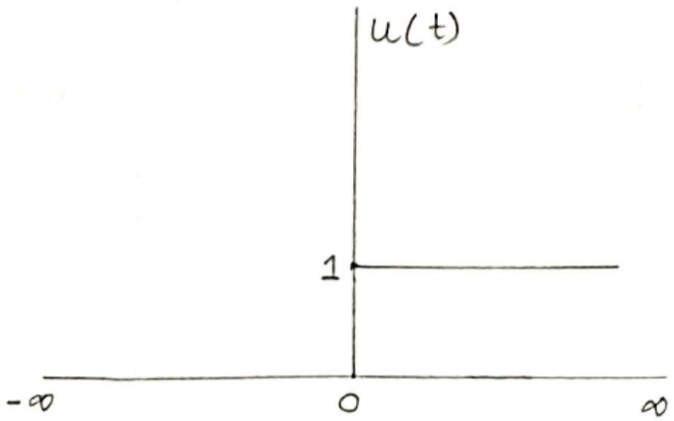
a. Unit step function →

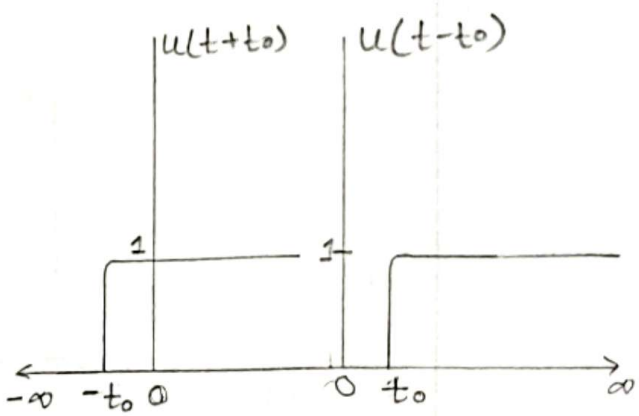
$$* u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

Signal shifting
1. Advance $(-t_0)$
2. Delay $(+t_0)$

If shifted,

$$* u(t - t_0) = \begin{cases} 1 & t > t_0 \\ 0 & t < t_0 \end{cases}$$





(Advance) (Delay)

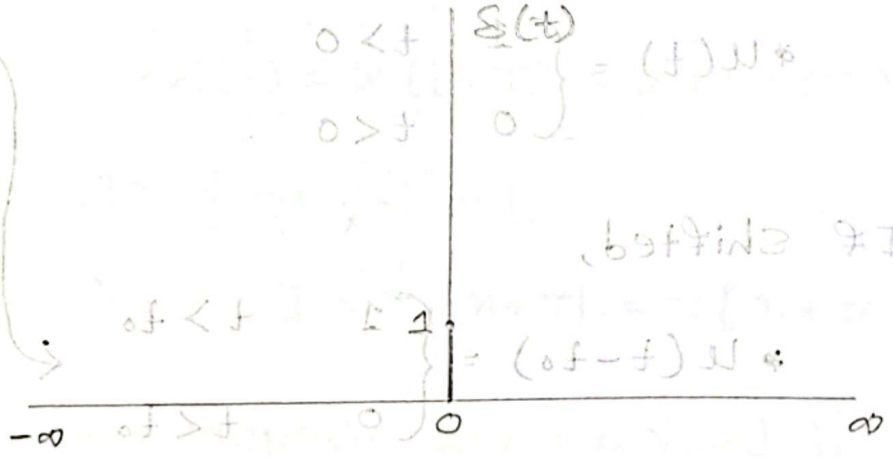
b. Unit impulse function →

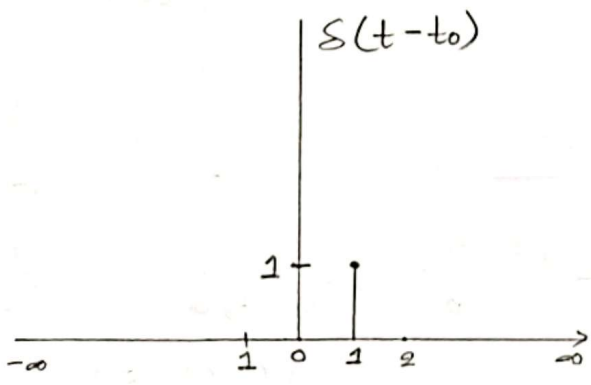
$$*\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & \text{otherwise} \end{cases}$$

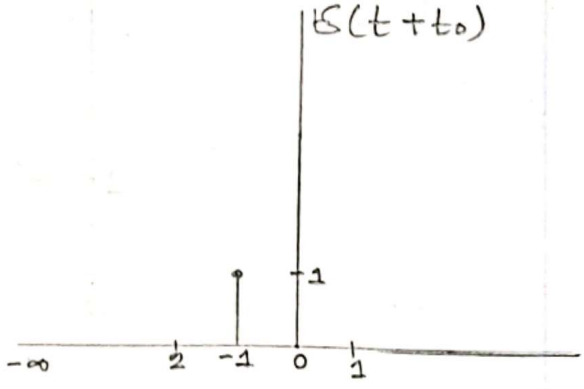
If shifted,

$$*\delta(t - t_0) = \begin{cases} 1 & t = t_0 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{cases} \text{Advance} = -t_0 \\ \text{Delay} = +t_0 \end{cases}$$

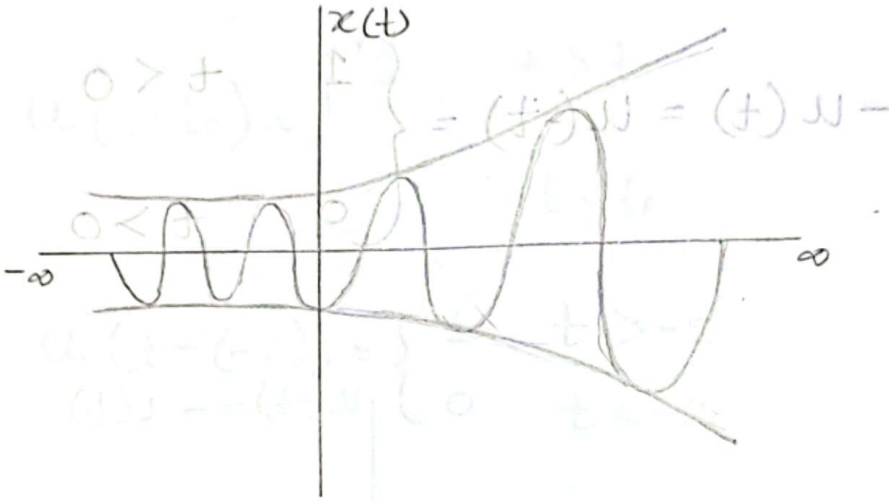




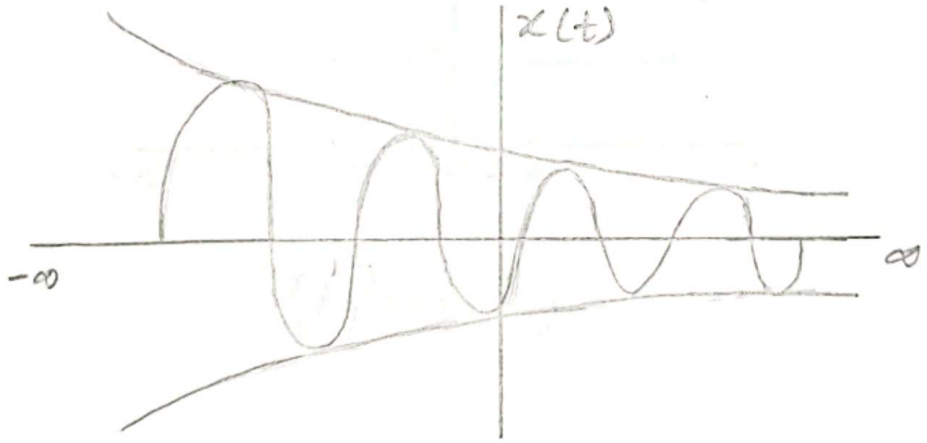


c. Complex exponential signals →

$x(t) = e^{j\omega t}$ → Increasing



$x(t) = e^{-j\omega t}$ → Decreasing



d. Sinusoidal signals →

$x(t) = A\cos(\omega t + \theta)$

$\omega = 2\pi f$; $f = \frac{1}{T}$;

A = Amplitude

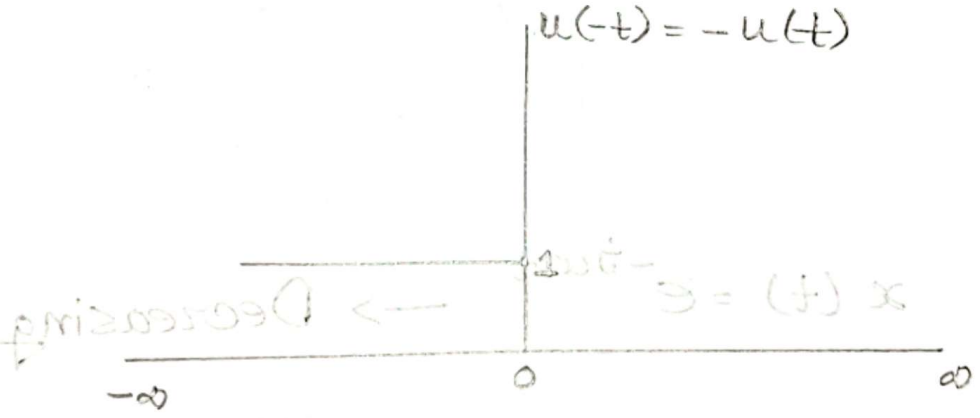
ω = Angular frequency

θ = Phase

Rotate / Flip / Fold —>

{Only for unit-step function}

$$-u(t) = u(-t) = \begin{cases} 1 & t < 0 \\ 0 & t > 0 \end{cases}$$



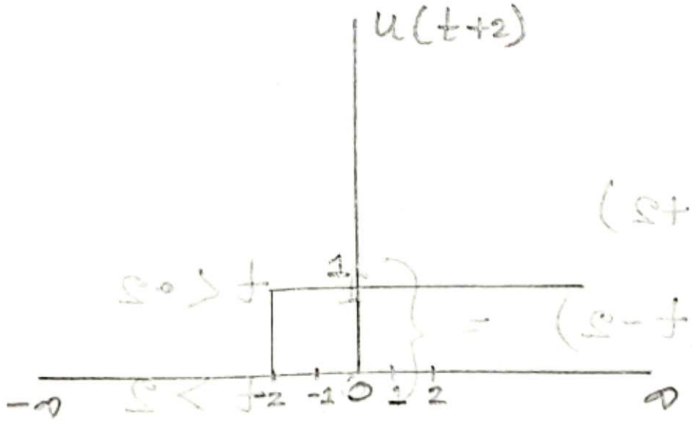
Problems

* $u(-t+2)$

$$u(t-t_0) = \begin{cases} 1 & t > t_0 \\ 0 & t < t_0 \end{cases}$$

$$u(t-(-2)) = \begin{cases} 1 & t > -2 \\ 0 & t < -2 \end{cases}$$

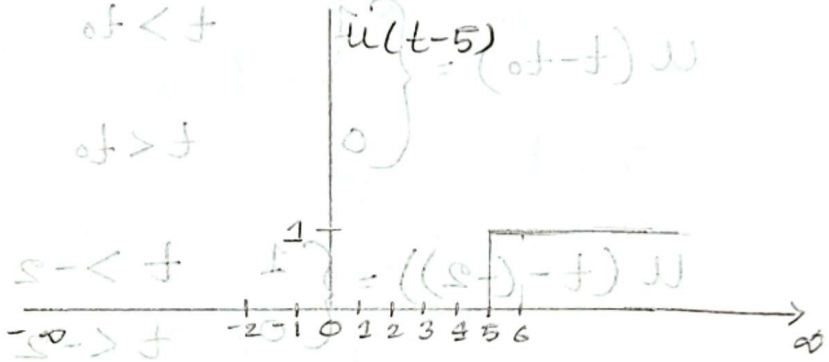
So,



(Advance)

* $u(t-5)$

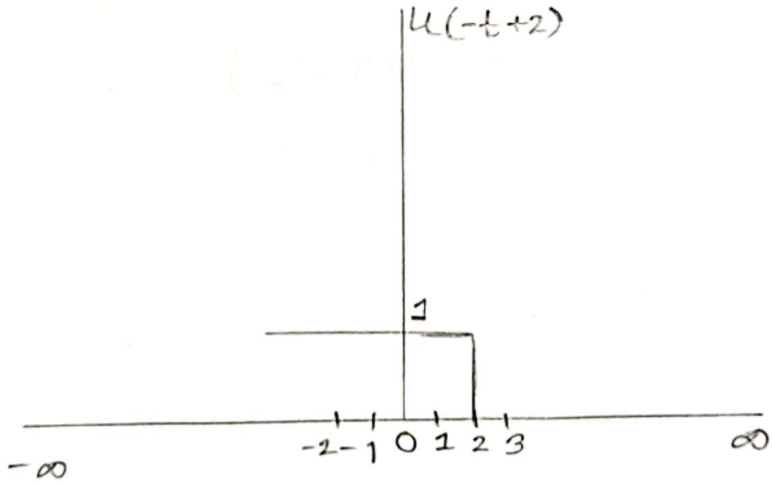
$$u(t-5) = \begin{cases} 1 & t > 5 \\ 0 & t < 5 \end{cases}$$



(Delay)

* $u(-t+2)$

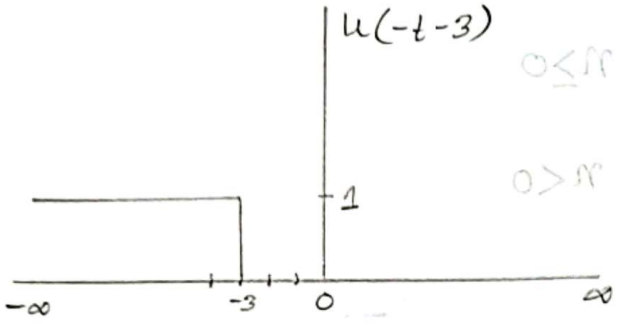
$$= -u(t-2) = \begin{cases} 1 & t < 2 \\ 0 & t > 2 \end{cases}$$



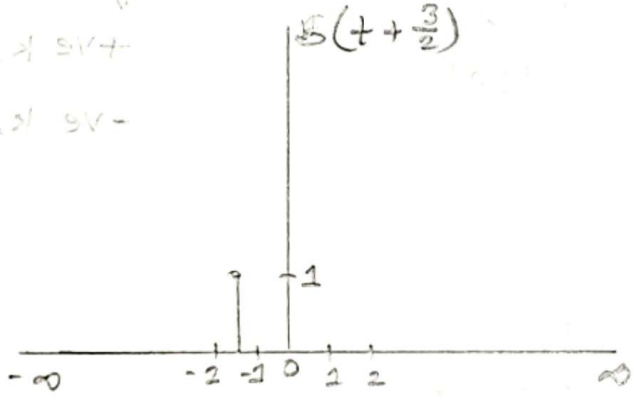
{Rotate / Flip}

* $u(-t-3)$

$$= -u(t+3) = -u(t-(-3)) = \begin{cases} 1 & t < -3 \\ 0 & t > -3 \end{cases}$$

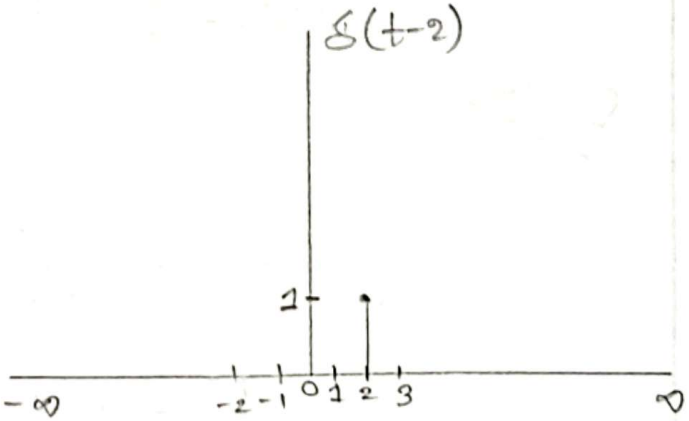


* $\delta\left(t+\frac{3}{2}\right) = \delta\left(t-(-1.5)\right) = \begin{cases} 1 & t = -1.5 \\ 0 & \text{otherwise} \end{cases}$



* $\delta(t-2)$

$$= \begin{cases} 1 & t = 2 \\ 0 & \text{otherwise} \end{cases}$$



Discrete time-signal

a. Unit step sequence →

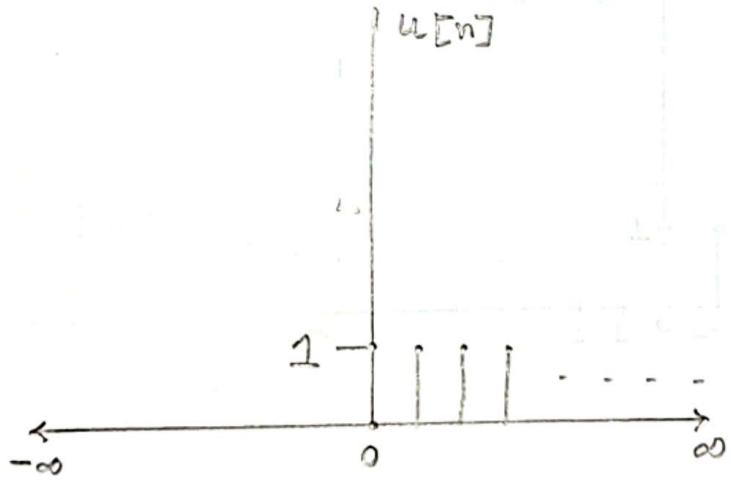
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

If shifted,

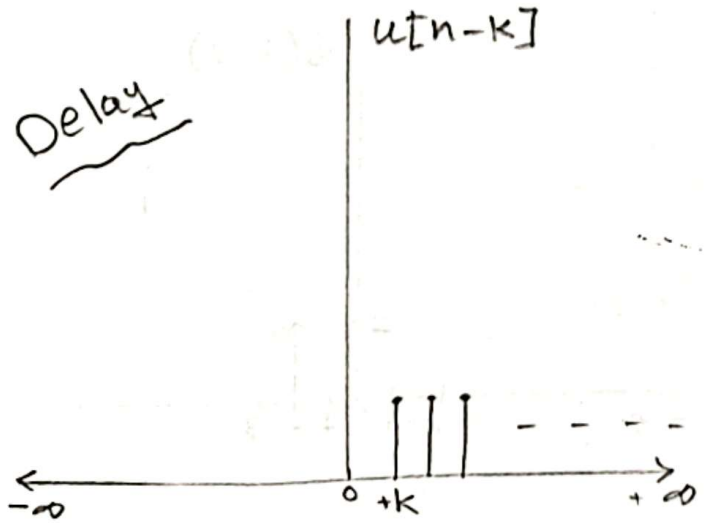
$$u[n-k] = \begin{cases} 1 & n \geq k \\ 0 & n < k \end{cases}$$

+ve k, Delay

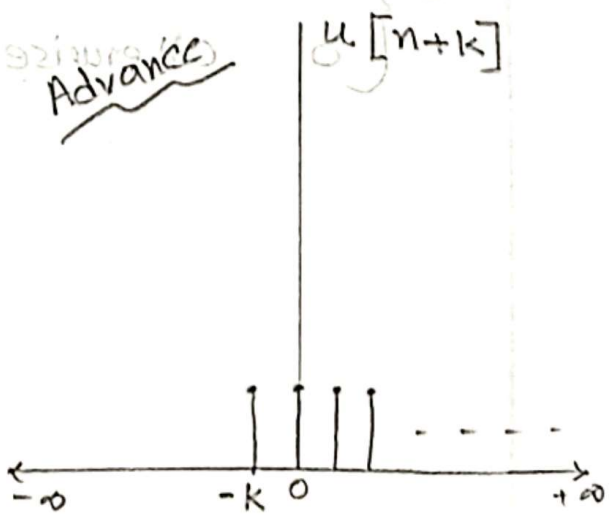
-ve k, Advance



Delay



Advance



b. Unit impulse sequence —

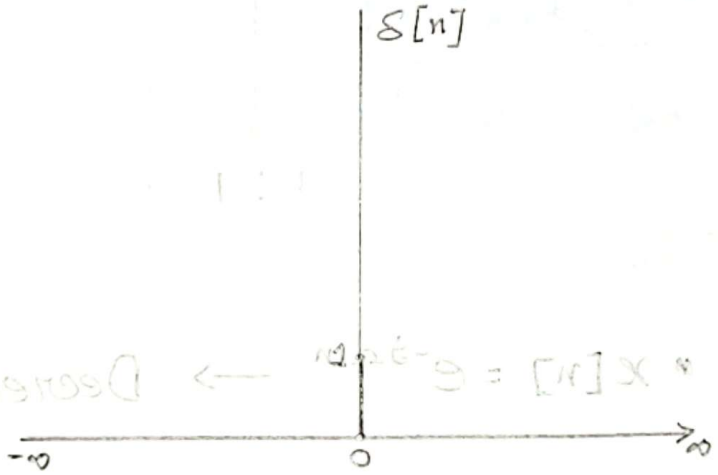
$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

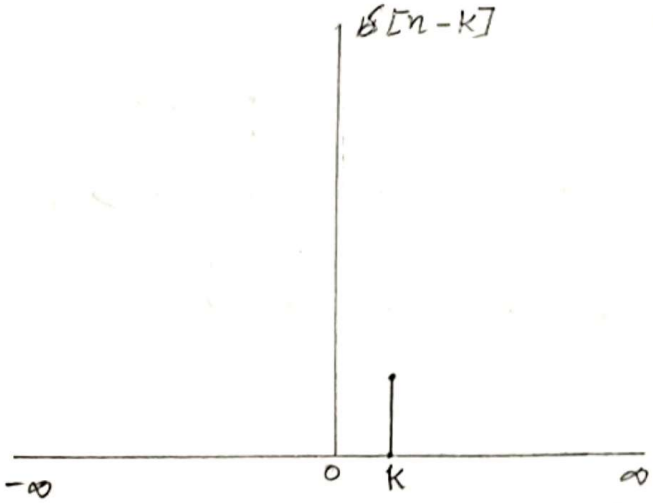
If shifted,

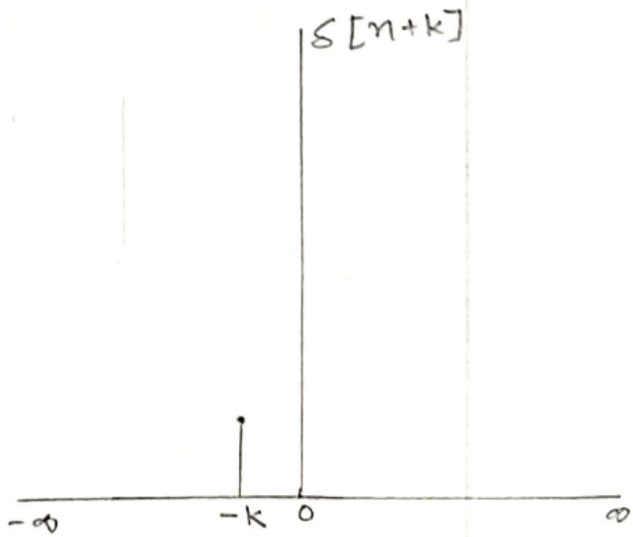
$$\delta[n-k] = \begin{cases} 1 & n = k \\ 0 & \text{otherwise} \end{cases}$$

→ +ve, k = delay

-ve, k = advance

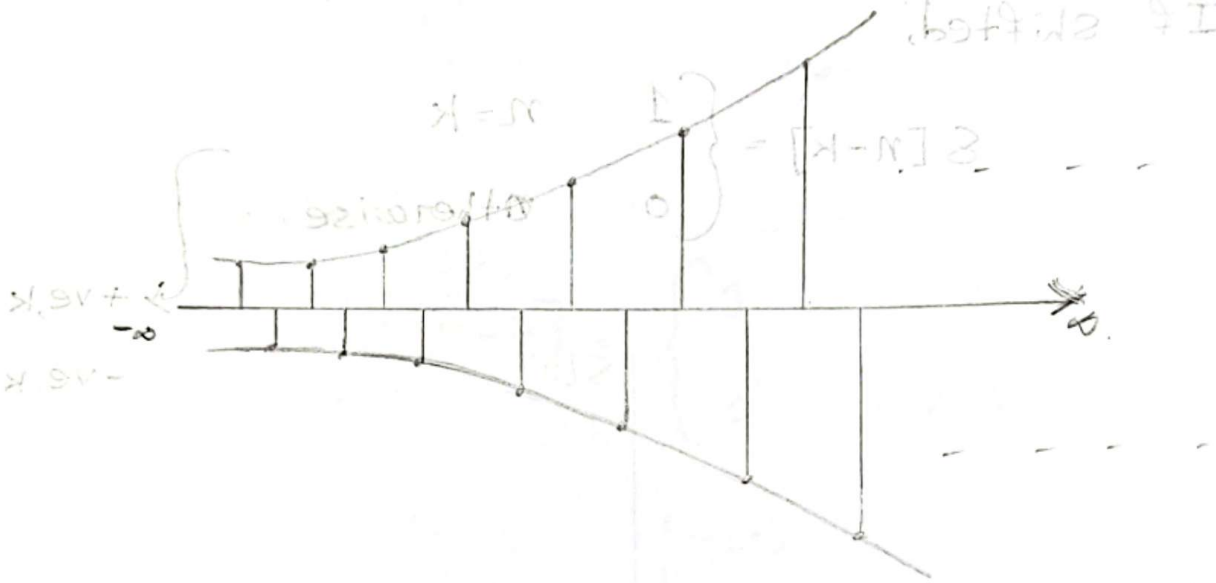




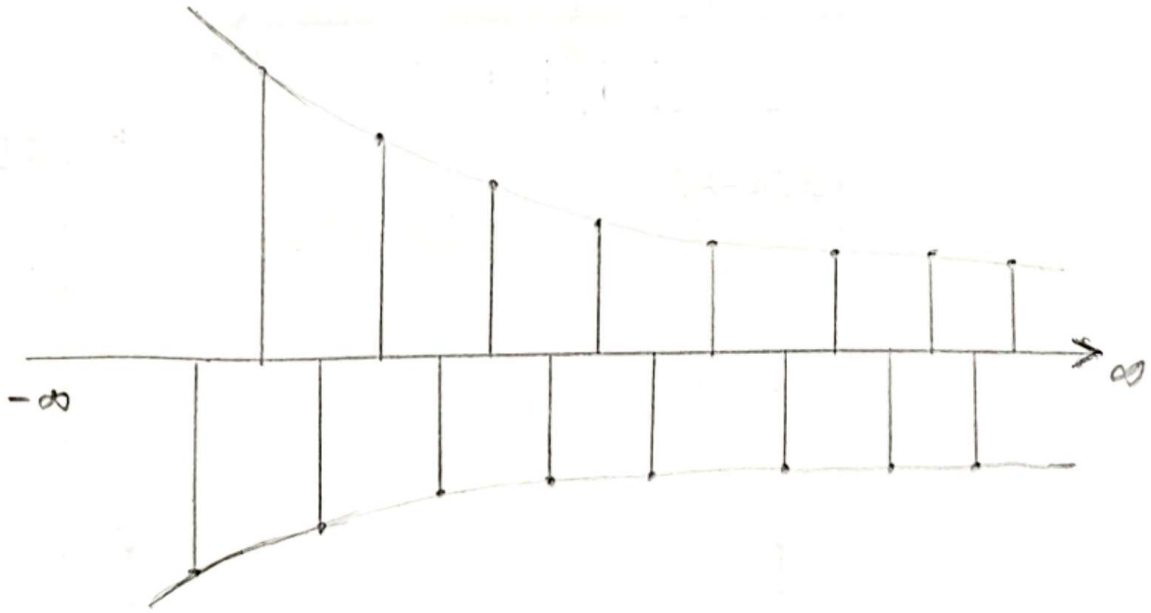


c. Complex exponential sequence ->

* $x[n] = e^{j\omega_0 n}$ -> Increasing



* $x[n] = e^{-j\omega_0 n}$ -> Decreasing



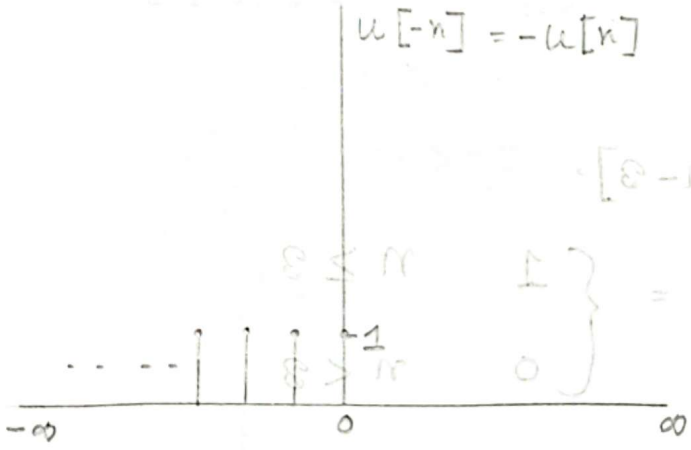
d. Sinusoidal sequence →

$$x[n] = A\cos(\Omega_0 n + \theta)$$

Rotate / Flip / Fold →

(Only for unit step sequence)

$$-u[n] = u[-n] = \begin{cases} 1 & n \le 0 \\ 0 & n > 0 \end{cases}$$

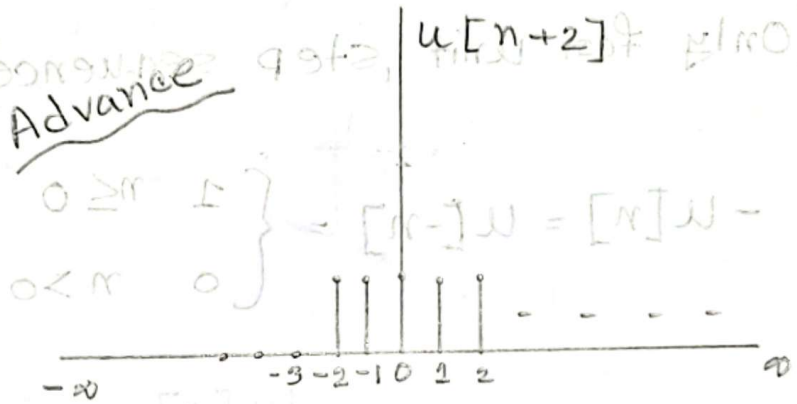


Can be shifted.

Problems

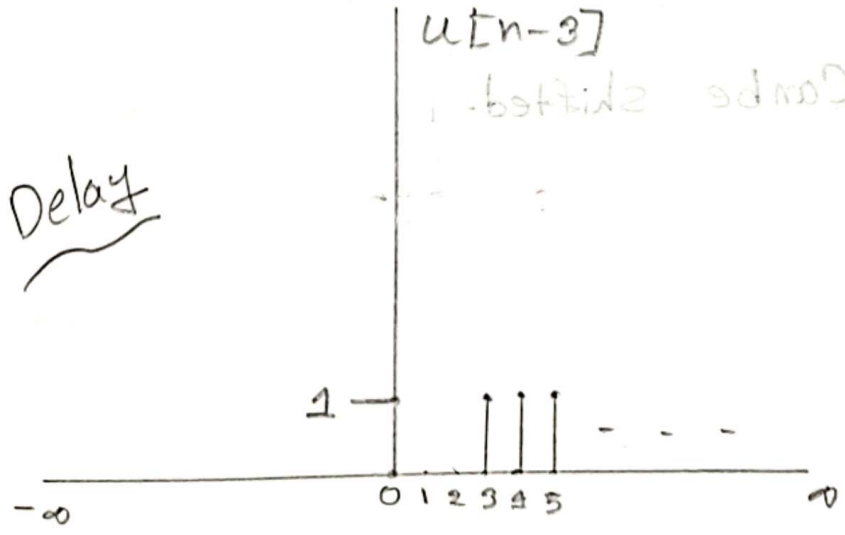
* $u[n+2]$

$= u[n-(-2)] = \begin{cases} 1 & n \geq -2 \\ 0 & n < -2 \end{cases}$



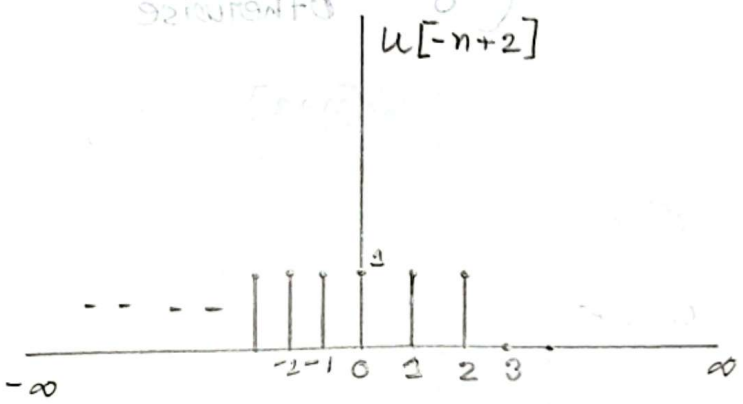
* $u[n-3]$

$= \begin{cases} 1 & n \geq 3 \\ 0 & n < 3 \end{cases}$



* $u[-n+2]$

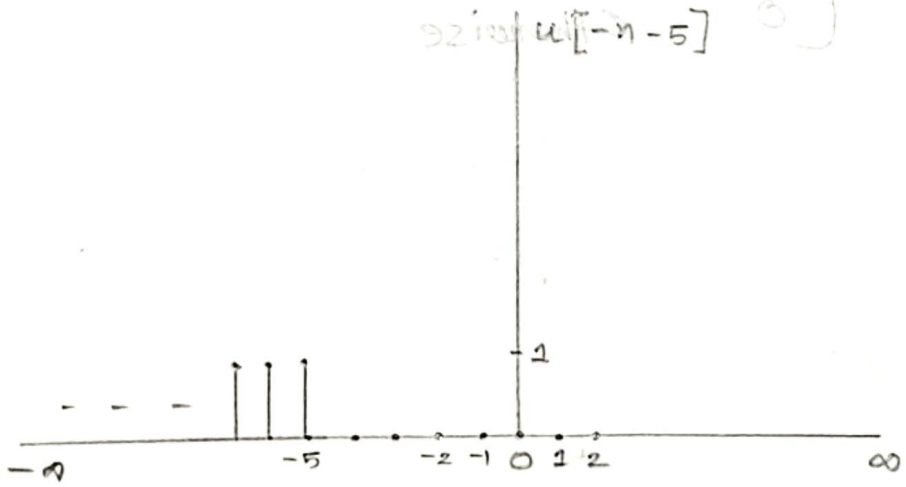
$= -u[n-2] = \begin{cases} 1 & n \leq 2 \\ 0 & n > 2 \end{cases}$



* $u[-n-5]$

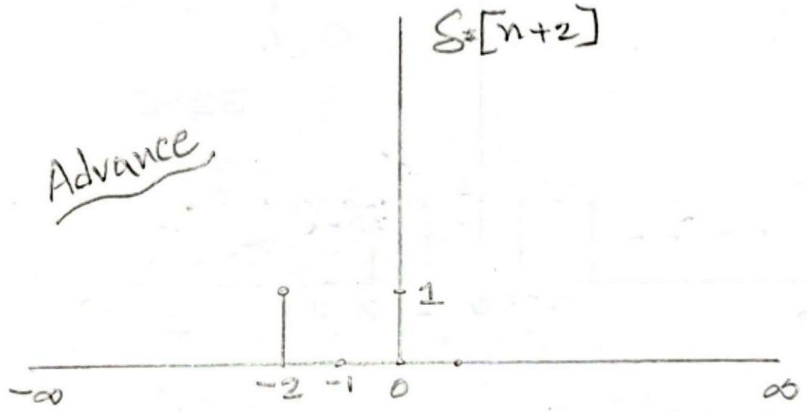
$= -u[n+5]$

$= -u[n+-(-5)] = \begin{cases} 1 & n \leq -5 \\ 0 & n > -5 \end{cases}$



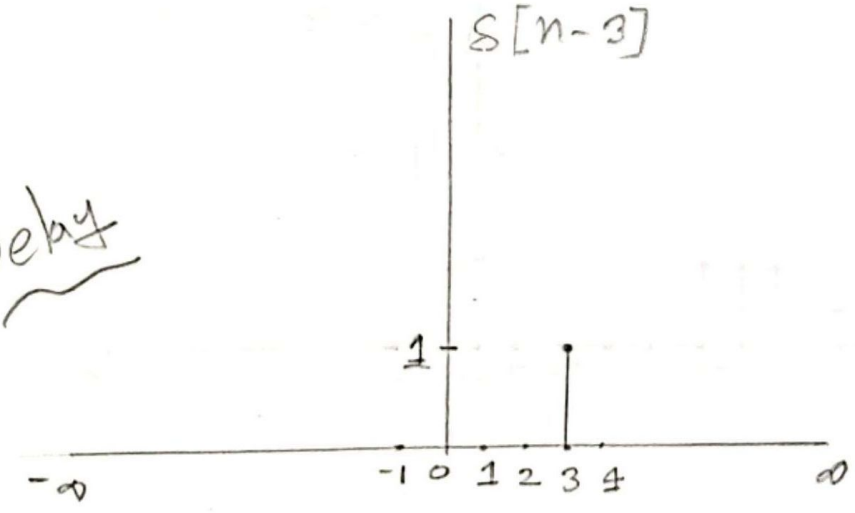
* $\delta[n+2]$

$$= \delta[n-(-2)] = \begin{cases} 1 & n = -2 \\ 0 & \text{otherwise} \end{cases}$$



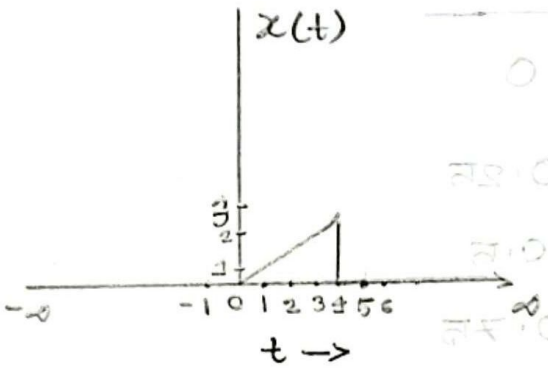
* $\delta[n-3]$

$$= \begin{cases} 1 & n = 3 \\ 0 & \text{otherwise} \end{cases}$$



Mathematical Problems ->

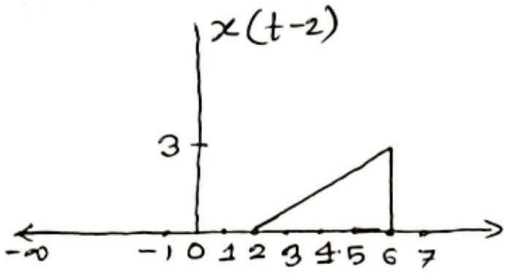
* Prob 1.1



t	x(t)
0	0
1	0.5
2	1
3	2
4	3
5	0
6	0
7	0

(From graph)

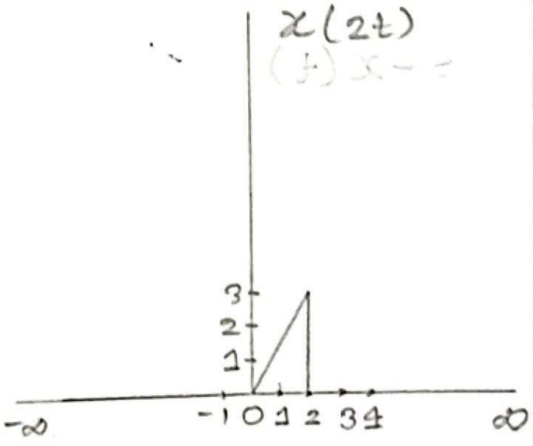
a) $x(t-2)$



b) $x(2t)$

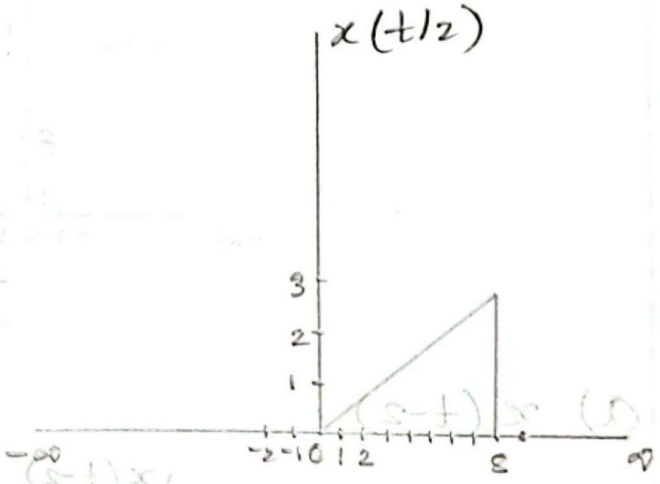
Now,

t	x(2t)
0	0
1	1
2	3
3	0
4	0



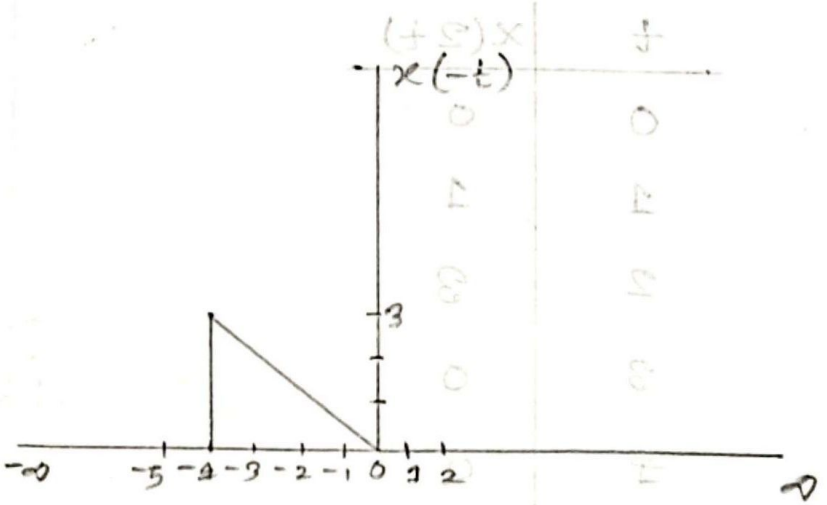
c) $x(t/2)$

t	$x(t/2)$
0	0
1	0.25
2	0.5
3	0.75
4	1
5	1.5
6	2
7	2.5
8	3
9	0

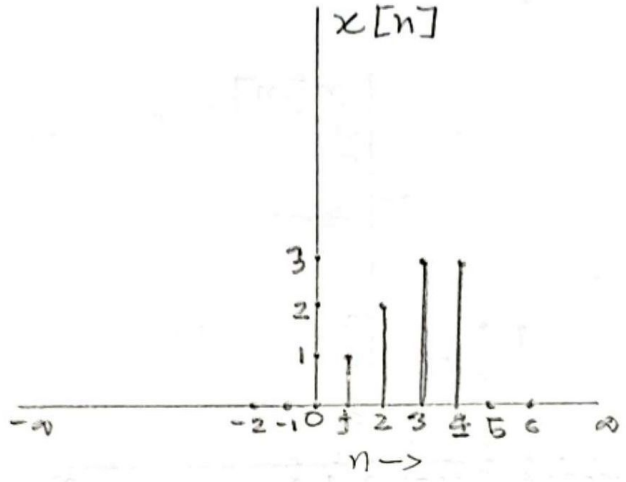


d) $x(-t)$

$= -x(t)$

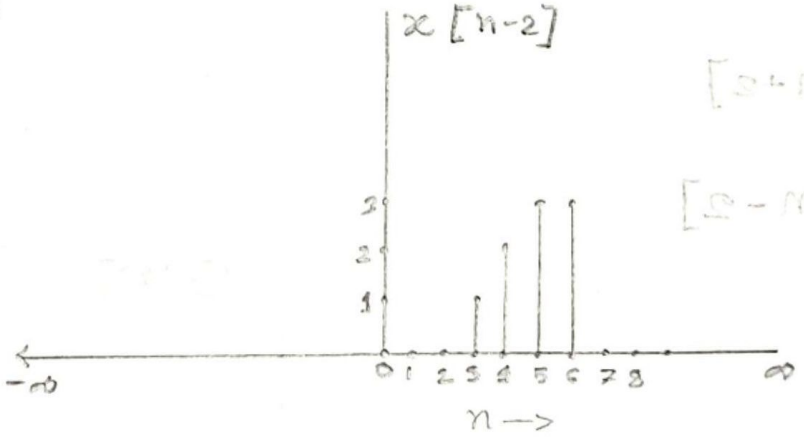


*Prob 1.2



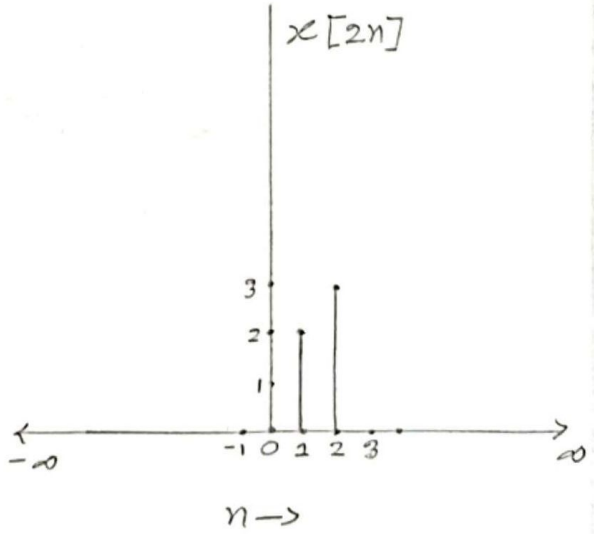
n	x[n]
0	0
1	1
2	2
3	3
4	3
5	0
6	0

a) $x[n-2]$



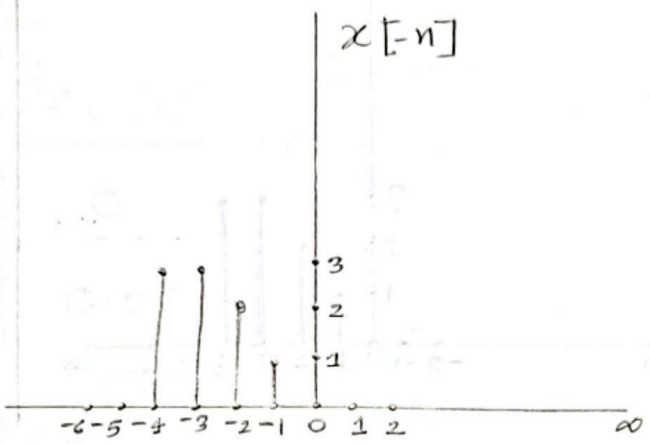
b) $x[2n]$

n	x[2n]
0	0
1	2
2	3
3	0



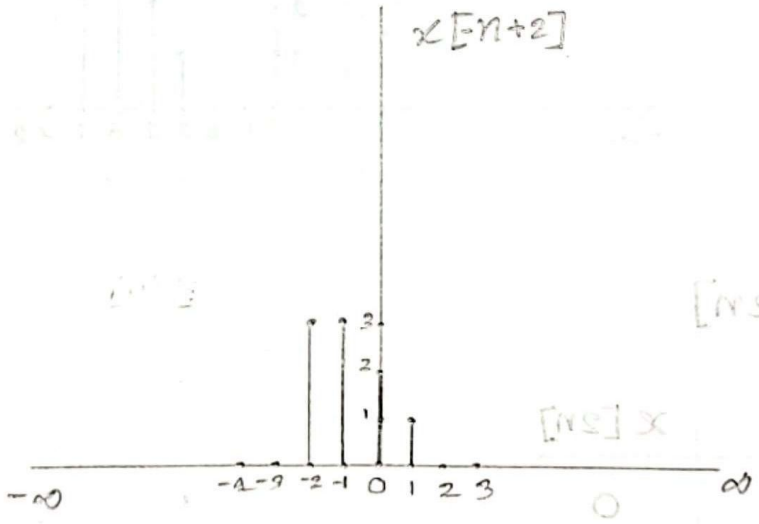
c) $x[-n]$

$= -x[n]$

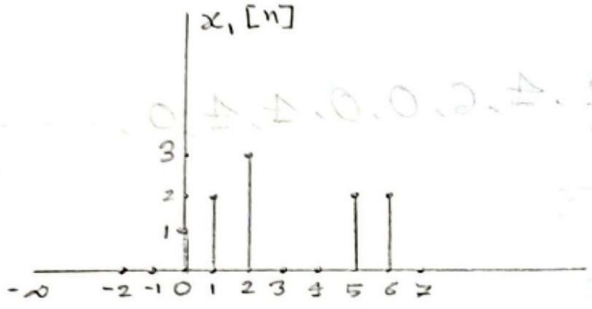


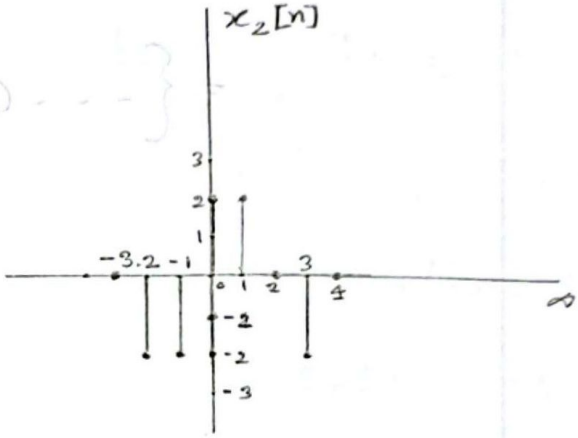
d) $x[-n+2]$

$= \div x[n-2]$



* Prob - 1.4





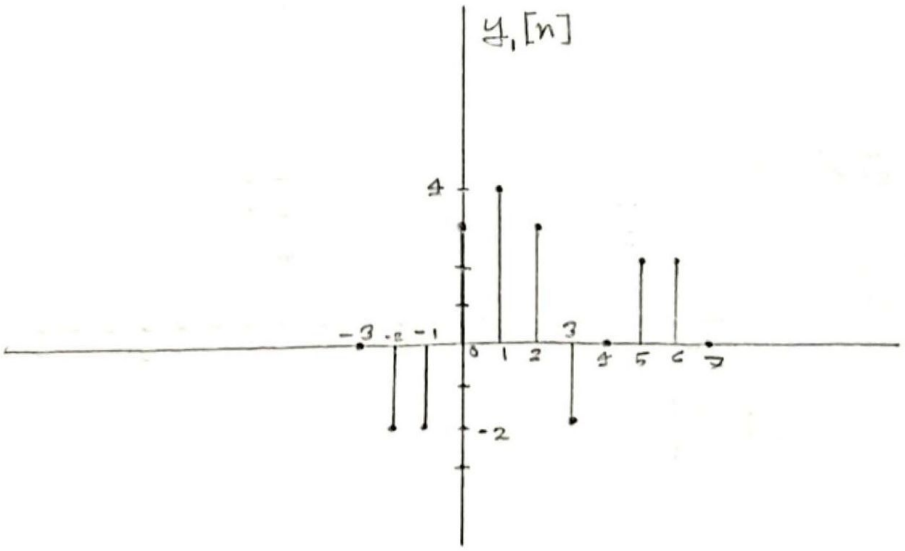
From graph,

$$x_1[n] = \{ - - -, -, 0, 0, \underset{\uparrow}{1}, 2, 3, 0, 0, 2, 2, 0 \ldots \}$$

$$x_2[n] = \{ - - -, 0, -2, -2, \underset{\uparrow}{2}, 2, 0, -2, 0, - - - \}$$

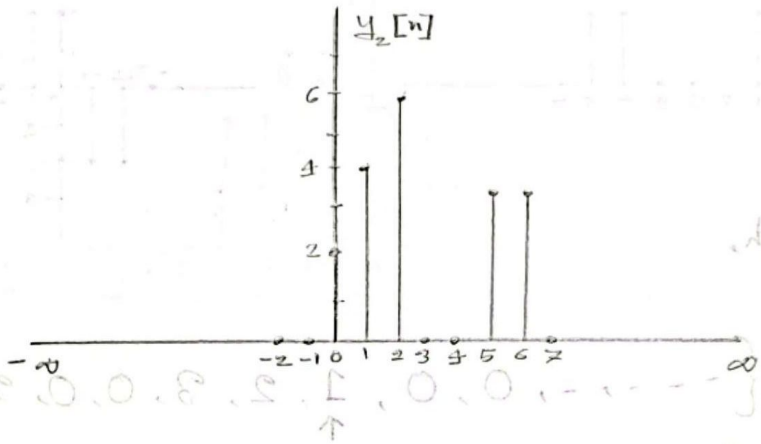
a) $y_1[n] = x_1[n] + x_2[n]$

$$= \{ - - -, 0, -2, -2, \underset{\uparrow}{3}, 4, 3, -2, 0, 2, 2, 0 \ldots \}$$



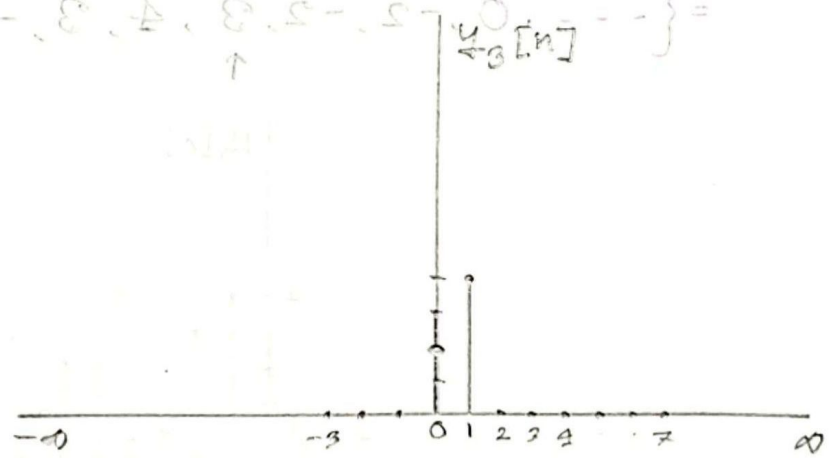
b) $y_2[n] = 2x_1[n]$

$$= \{\ldots 0, 0, \underset{\uparrow}{2}, 4, 6, 0, 0, 4, 4, 0 \ldots\}$$



c) $y_3[n] = x_1[n] \cdot x_2[n]$

$$= \{\ldots 0, 0, 0, \underset{\uparrow}{2}, 4, 0, 0, 0, 0, 0, 0 \ldots\}$$



* Prob 1.14-

Let, $x_1(t)$ and $x_2(t)$ be periodic signals with fundamental periods T_1 and T_2, respectively. Under what condition is the sum $x(t) = x_1(t) + x_2(t)$ periodic and what is the fundamental period of $x(t)$ if it is periodic?

=>

Since,

$x_1(t)$ & $x_2(t)$ are periodic,

$$\therefore x_1(t) = x_1(t+T_1) = x_1(t+mT_1) \quad [m = +ve]$$

$$x_2(t) = x_2(t+T_2) = x_2(t+kT_2) \quad [k = +ve]$$

Coditionally,

$$x(t) = x_1(t) + x_2(t)$$

$$\Rightarrow x(t) = x_1(t+mT_1) + x_2(t+kT_2)$$

$$\Rightarrow x(t+T) = x_1(t+mT_1) + x_2(t+kT_2)$$

So, $T = mT_1 = kT_2$

$$\Rightarrow mT_1 = kT_2$$

$$\therefore \frac{T_1}{T_2} = \frac{k}{m} = \text{Rational number}$$

(যুক্তি সংখ্যা)

∴ Fundamental period of time, L.C.M of k and m.

Ans

* Prob 1.16 -

$x(t) = A\cos(\omega t + \theta)$; $x[n] = A\cos(\Omega n + \theta)$

a) $x(t) = \cos\left(t + \frac{\pi}{4}\right)$

Here,

$\omega = 1$; $\theta = \frac{\pi}{4}$

$\omega = 1$

$\Rightarrow 2\pi f = 1$

$\Rightarrow f = \frac{1}{2\pi}$ $\left[T = \frac{1}{f}\right]$

$\therefore T = 2\pi$ (Ans)

b) $x(t) = \sin\frac{2\pi}{3}t$

Here,

$\omega = \frac{2\pi}{3}$

$\Rightarrow 2\pi f = \frac{2\pi}{3}$

$\Rightarrow f = \frac{1}{3}$

$\therefore T = 3$ (Ans)

c) $x(t) = \cos\frac{\pi}{3}t + \sin\frac{\pi}{4}t$

Here,

$x_1(t) = \cos\frac{\pi}{3}t$

$\omega_1 = \frac{\pi}{3}$

$\Rightarrow 2\pi f_1 = \frac{\pi}{3}$

$\Rightarrow f_1 = \frac{1}{6}$

$\therefore T_1 = 6$

$x_2(t) = \sin\frac{\pi}{4}t$

$\omega_2 = \frac{\pi}{4}$

$\Rightarrow 2\pi f_2 = \frac{\pi}{4}$

$\Rightarrow f_2 = \frac{1}{8}$

$\therefore T_2 = 8$

~~$\therefore$ Fundamental~~

Now,

$$\frac{T_1}{T_2} = \frac{6}{8} = \frac{3}{4} = \text{Rational.}$$

So, Fundamental time period,

LCM of T_1, T_2

$$2 \mid 6, 8$$
$$3, 4$$

$\therefore T = 2 \times 3 \times 4 = 24$

(Ans)

d) $x(t) = \cos t + \sin\sqrt{2}t$

Here,

$x_1(t) = \cos t$

$\omega_1 = 1$

$\Rightarrow 2\pi f_1 = 1$

$\Rightarrow f_1 = \frac{1}{2\pi}$

$\therefore T_1 = 2\pi$

$x_2(t) = \sin\sqrt{2}t$

$\omega_2 = \sqrt{2}$

$\Rightarrow 2\pi f_2 = \sqrt{2}$

$\Rightarrow f_2 = \frac{\sqrt{2}}{2\pi}$

$\therefore T_2 = \frac{2\pi}{\sqrt{2}}$

Now,

$$\frac{T_1}{T_2} = \frac{2\pi}{\frac{2\pi}{\sqrt{2}}} = \sqrt{2} = \text{Irrational}.$$

So,

$x(t)$ is not periodic.

e) $x(t) = \sin^2 t$

$= \frac{1}{2}(1 - \cos 2t)$

$= \frac{1}{2} - \frac{1}{2}\cos 2t$

Here,

$x_1(t) = \frac{1}{2}$; constant means periodic (arbitary).

$x_2(t) = -\frac{1}{2}\cos 2t$

$\omega_2 = 2$

$\Rightarrow 2\pi f_2 = 2$

$\Rightarrow f_2 = \frac{2}{2\pi}$

$\therefore T_2 = \pi$

$\therefore$ Fundamental period $= \pi$.

f) $x(t) = e^{j[(\pi/2)t - 1]}$

$= e^{j(\pi/2)t} \cdot e^{-j}$

$\therefore \omega = \frac{\pi}{2}$

$\Rightarrow 2\pi f = \frac{\pi}{2}$

$\therefore T = 4$

Ans) Fundamental period, $T = 4$.

g) $x[n] = e^{j(\pi/4)n}$

Here, [Discrete]

$$\Omega = \frac{\pi}{4}$$

$$\Rightarrow 2\pi f = \frac{\pi}{4} \Rightarrow f = \frac{1}{8}$$

$$\therefore N = 8$$

∴ Fundamental time period, $N = 8$.

h) $x[n] = \cos \frac{1}{4} n$

Here,

$$\Omega = \frac{1}{4}$$

$$\Rightarrow 2\pi f = \frac{1}{4}$$

$$\Rightarrow f = \frac{1}{8\pi}$$

~~$\therefore N = 8\pi$~~

Here, f = irrational number.

So the signal is nonperiodic.

* ($\frac{1}{\pi}$ = Irrational number).

i) $x[n] = \cos\frac{\pi}{3}n + \sin\frac{\pi}{4}n$

Here,

$x_1[n] = \cos\frac{\pi}{3}n$ $\qquad$ $x_2[n] = \sin\frac{\pi}{4}n$

$\Omega_1 = \frac{\pi}{3}$ $\qquad$ $\Omega_2 = \frac{\pi}{4}$

$\Rightarrow 2\pi f_1 = \frac{\pi}{3}$ $\qquad$ $\Rightarrow 2\pi f_2 = \frac{\pi}{4}$

$\Rightarrow f_1 = \frac{1}{6}$ $\qquad$ $\Rightarrow f_2 = \frac{1}{8}$

$\therefore N_1 = 6$ $\qquad$ $\therefore N_2 = 8$

Now,

$$\frac{N_1}{N_2} = \frac{6}{8} = \frac{3}{4} = \text{Rational number.}$$

So, Fundamental time period,

LCM of N_1, N_2

$$2 \mid 6, 8$$
$$3, 4$$

$\therefore N = 2 \times 3 \times 4 = 24$

(Ans)

j) $x[n] = \cos^2 \frac{\pi}{8} n$

$$= \frac{1}{2}\left(1 + \cos 2 \times \frac{\pi}{8} n\right)$$

$$= \frac{1}{2} + \frac{1}{2} \cos \frac{\pi}{4} n$$

Here,

$x_1[n] = \frac{1}{2}$; constant means periodic.

$x_2[n] = \frac{1}{2} \cos \frac{\pi}{4} n$

Here,

$\Omega_2 = \frac{\pi}{4}$

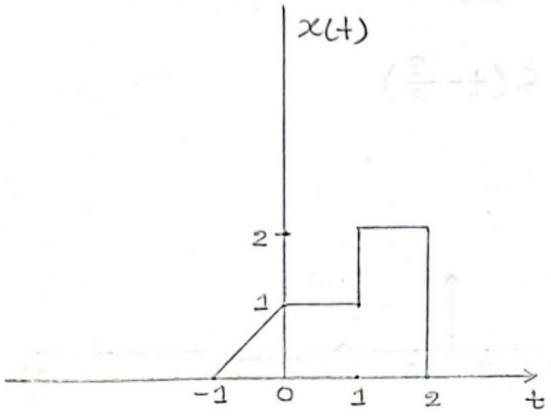
$\Rightarrow 2\pi f_2 = \frac{\pi}{4}$

$\Rightarrow f_2 = \frac{1}{8}$

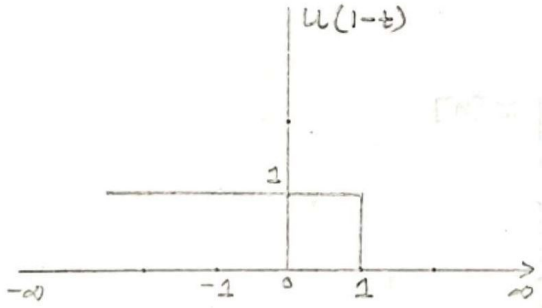
$\therefore N_2 = 8$

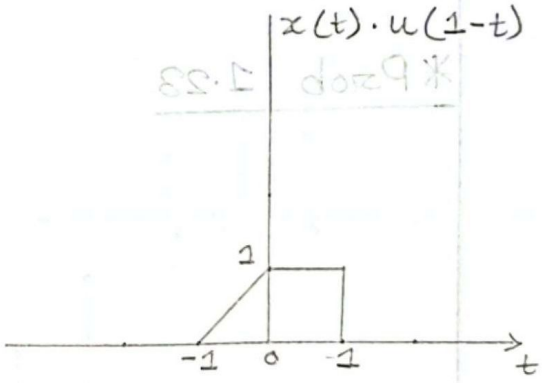
So fundamental time period, $N = 8$ (Ans)

* Prob 1.22

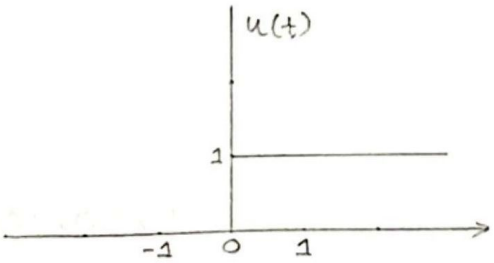


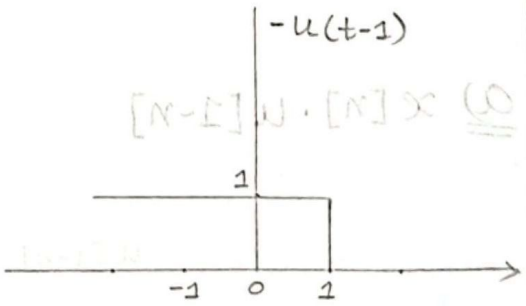
a) $x(t) \cdot u(1-t)$

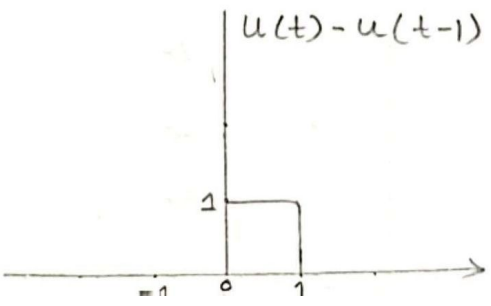


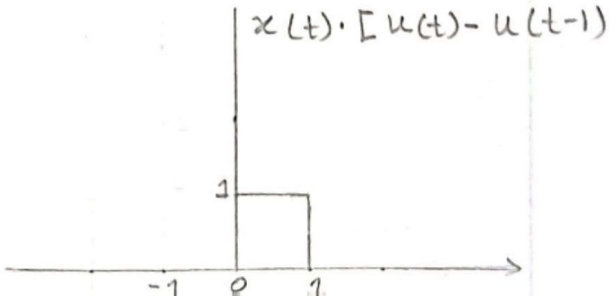


b) $x(t) \cdot [u(t) - u(t-1)]$

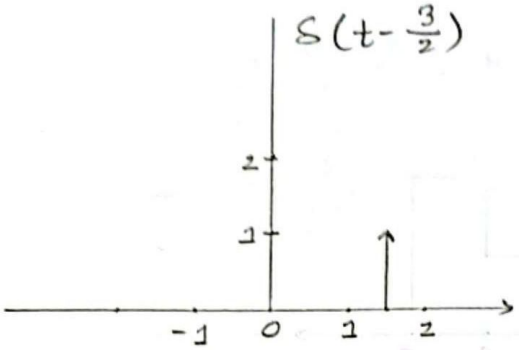


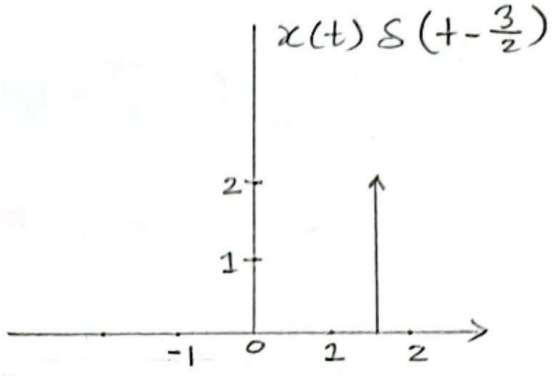




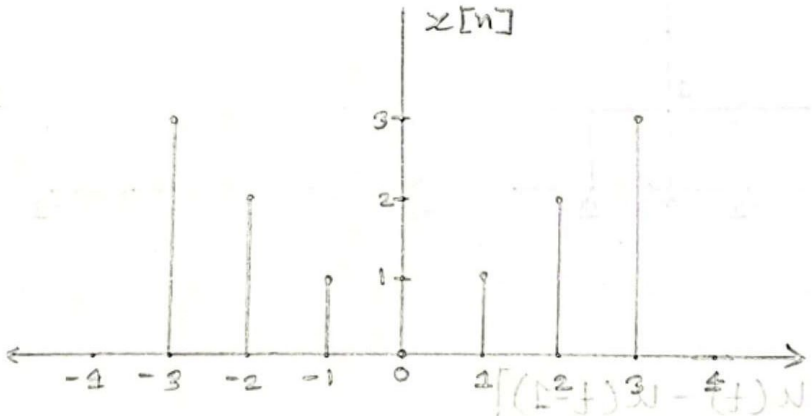


c) $x(t)\,\delta(t-\frac{3}{2})$

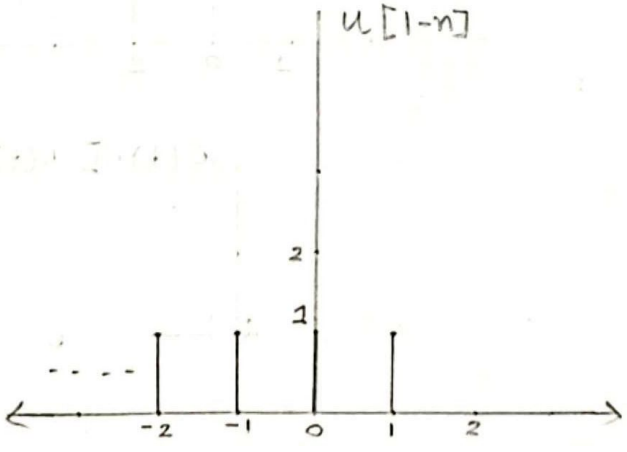


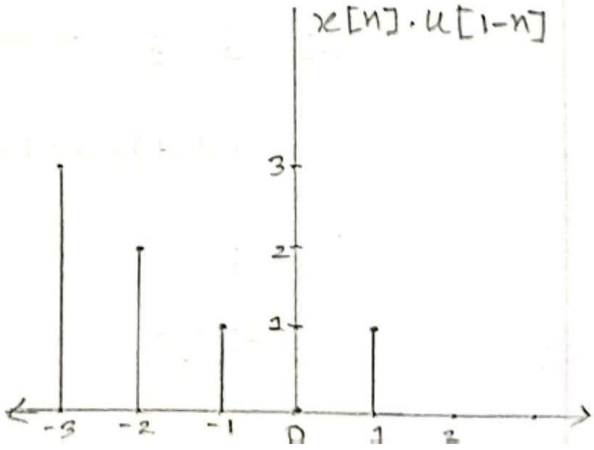


*Prob 1.23

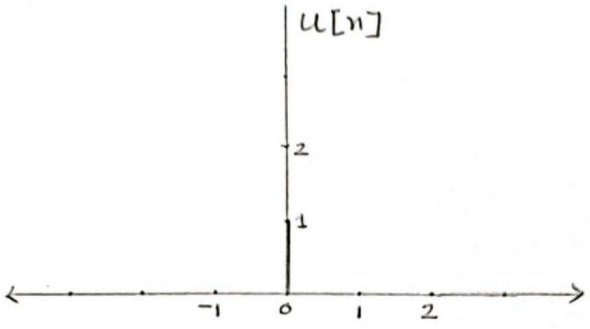


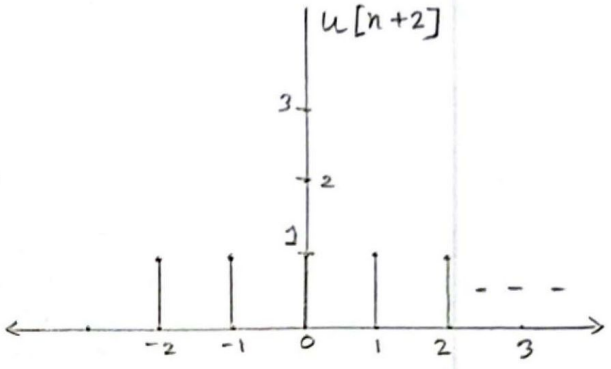
a) $x[n]\cdot u[1-n]$

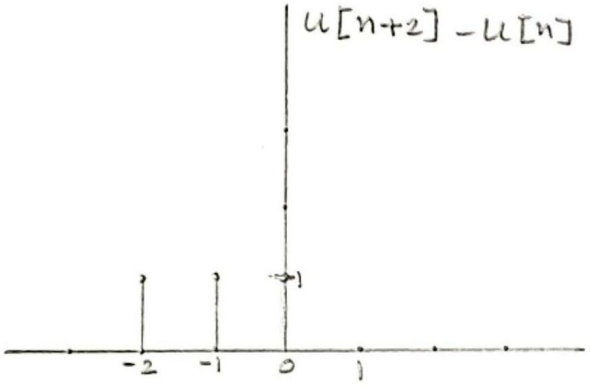


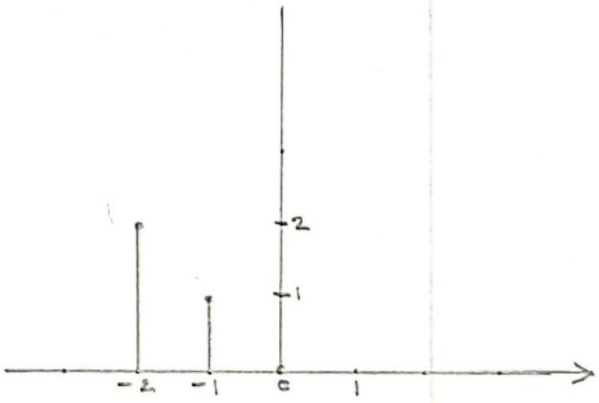


b) $x[n] \cdot \{u[n+2] - u[n]\}$

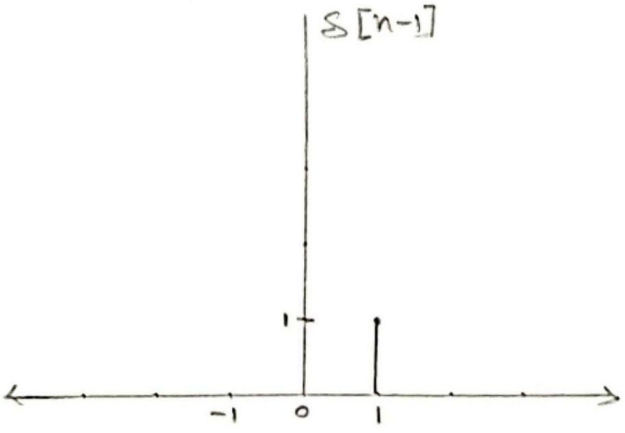


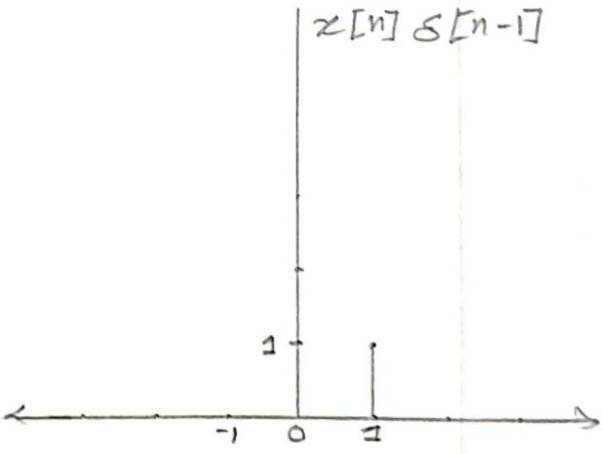






c) $x[n]\,\delta[n-1]$

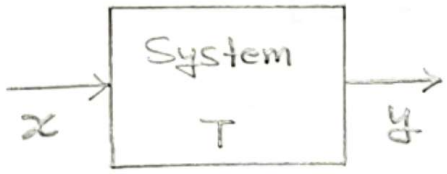




System →

System is a mathematical model that relates the input signal to output signal.

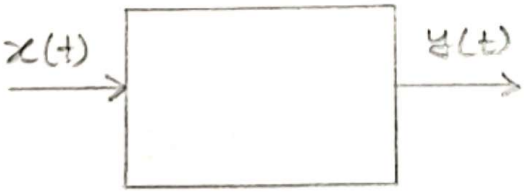
$$y = Tx = T\{x\}$$



Classification →

A. Continious time system

Input and output both are continious.



B. Discrete time system

Input and output both are discrete.



C. System without memory → (Memoryless)

Input, output both are present time.

D. System with memory →

Present input → Future output

Past input → Present output

E. Causal and non-causal system →

Present input → Present output

Past input → Present output

*** # All memoryless systems are causal but all causal systems are not memoryless. (vice-versa)

=>

A memoryless system in which present output depends on present input. But in a causal system present output depends on present input or on past input. Memoryless system doesn't satisfy the condition of past input for a present output. That's why all memoryless systems are causal but all causal systems are not memoryless.

F. Linear and non-linear systems →

For linear system →

{ Condition 1 → Additivity
{ Condition 2 → Scaling / Homogenity

1. Additivity →



$$T\{x_1+x_2\} = y_1+y_2$$

2. Scaling →



$$T\{ax\} = ay$$

Linear combition condition →

* $T\{a_1x_1 + a_2x_2\} = a_1y_1 + a_2y_2$

Non-linear → System which doesn't satisfy the condition of linear system.

F. Time invarient →

Time shifting (delay or advance) in output, will be same as in input.

$T\{x(t-t_0)\} = y(t-t_0)$ → Continious

$T\{x[n-k]\} = y[n-k]$ → Discrete

G. Time varying →

System that don't follow the invarient condition.

H. Linear time-invarient system (LTI) →

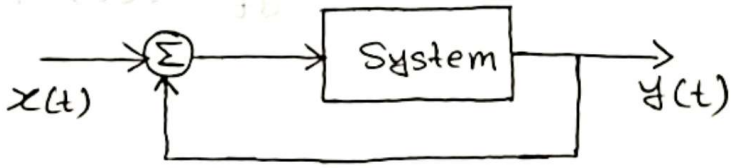
Linear system + Time invarient system

(Condition)

I. Stable system →

Bounded input / Bounded output

J. Feedback system →

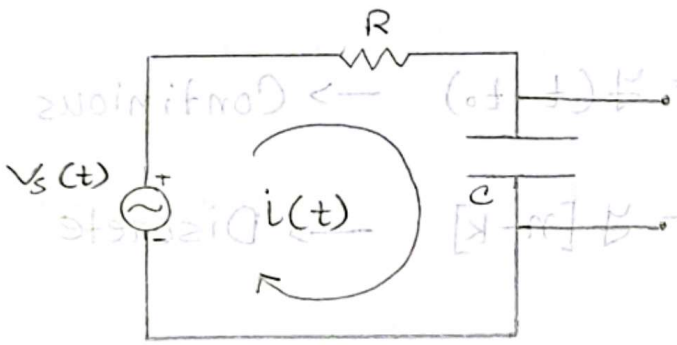


Capacitor formula → 1. $i(t) = C \frac{d}{dt} V_c(t)$

2. $V_c(t) = \frac{1}{C} \int i(t) \cdot dt$

Mathematical problems →

* Prob 1.32



Find relationtion between the input $x(t)$ and the output $y(t)$.

a) If $x(t) = V_s(t)$ and $y(t) = V_c(t)$.

=> KVL,

$$V_s(t) = V_R(t) + V_c(t)$$

$$\Rightarrow V_s(t) = R * i(t) + V_c(t)$$

$$\Rightarrow x(t) = R * i(t) + y(t)$$

$$\Rightarrow x(t) = RC \frac{d}{dt} V_c(t) + y(t)$$

$$\Rightarrow x(t) = RC \frac{d}{dt} y(t) + y(t)$$

(Ans)

b) If $x(t) = V_s(t)$ and $y(t) = i(t)$.

=>

KVL,

$$V_s(t) = V_R(t) + V_c(t)$$

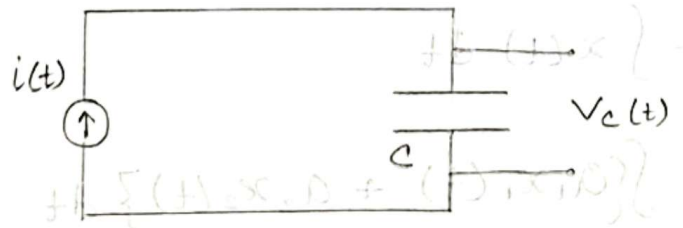
$$\Rightarrow V_s(t) = R \times i(t) + V_c(t)$$

$$\Rightarrow x(t) = R\,y(t) + \frac{1}{C}\int i(t)\cdot dt$$

$$\Rightarrow x(t) = R\,y(t) + \frac{1}{C}\int y(t)\cdot dt$$

(Ans)

Prob 1.33



Let, input $x(t) = i(t)$

output $y(t) = V_c(t)$

a) Find the input-output relationship.

=> $V_c(t) = V_c(t)$

$$\Rightarrow y(t) = \frac{1}{C}\int i(t)\cdot dt = \frac{1}{C}\int x(t)\cdot dt$$

(Ans)

b) Determine —>

1) Memoryless —>

Not memoryless, because capacitor stores past valus for present output.

11) Causal —>

Yes causal, because the system gets present output for past input values.

111) Linear —>

Let,

$$x(t) = a_1 x_1(t) + a_2 x_2(t) \quad \text{[Condition]}$$

$$y(t) = \frac{1}{c} \int x(t)\, dt$$

$$= \frac{1}{c} \int \{a_1 x_1(t) + a_2 x_2(t)\}\, dt$$

$$= \frac{1}{c} \int a_1 x_1(t)\, dt + \frac{1}{c} \int a_2 x_2(t)\, dt$$

$$= a_1 \frac{1}{c} \int x_1(t)\, dt + a_2 \frac{1}{c} \int x_2(t)\, dt$$

$$= a_1 y_1(t) + a_2 y_2(t)$$

So, the system is linear.

iv) Time - invarient →

Condition,

$$T\{x(t-t_0)\} = y(t-t_0)$$

$$y(t) = \frac{1}{c}\int x(t)\,dt$$

$$= \frac{1}{c}\int x(t-t_0)\,dt \qquad \text{let, } t-t_0 = p$$

$$= \frac{1}{c}\int x(p)\cdot dp$$

$$= y(p)$$

$$= y(t-t_0)$$

So, the system is time-invarient.

*Prob 1.36

Given, $y[n] = T\{x[n]\} = x[n-1]$; Determine.

a) Not memoryless, because the present output n, depends on the past input $n-1$ (delay)

b) Causal, because the output is present and the input is past(delay).

c) Let,

$$x[n] = a_1 x_1[n] + a_2 x_2[n]$$

Given,

$$y[n] = x[n-1]$$

$$= a_1 x_1[n-1] + a_2 x_2[n-1]$$

$$= a_1 y_1[n] + a_2 y_2[n]$$

So, the system is linear.

d) Let,

$$T\{x[n-k]\} = y[n-k] \qquad \text{[condition]}$$

Given,

$$y[n] = x[n-1] \quad ;\therefore y[n-k] = x[n-k-1]$$

~~$T\{x[n-k-1]\} =$~~

$$y[n-k] = x[n-k-1]$$

let,

$$n-k = p$$

$$= x[p-1]$$

$$= y[p]$$

$$= y[n-k]$$

So, the system is time-invarient.

* Prob 1.38

$y[n] = T\{x[n]\} = nx[n]$: Determine.

$\Rightarrow$

a) Memoryless, because the present output n depends on the ~~past~~ input n (present).

b) Causal, because the present output is same as present input.

c) Condition,

$$x[n] = a_1 x_1[n] + a_2 x_2[n]$$

Given,

$$y[n] = nx[n] = n\{a_1 x_1[n] + a_2 x_2[n]\}$$

$$= n a_1 x_1[n] + n a_2 x_2[n]$$

(Linear)

$$= a_1 n x_1[n] + a_2 n x_2[n]$$

$$= a_1 y_1[n] + a_2 y_2[n]$$

d) ~~Put~~, Given,

$$y[n] = n x[n] \quad \therefore \text{ let, } n = n-k$$

$$\therefore y[n-k] = (n-k)\, x[n-k]$$

$$= n x[n-k] - k x[n-k]$$

So, the system is not time invarient.